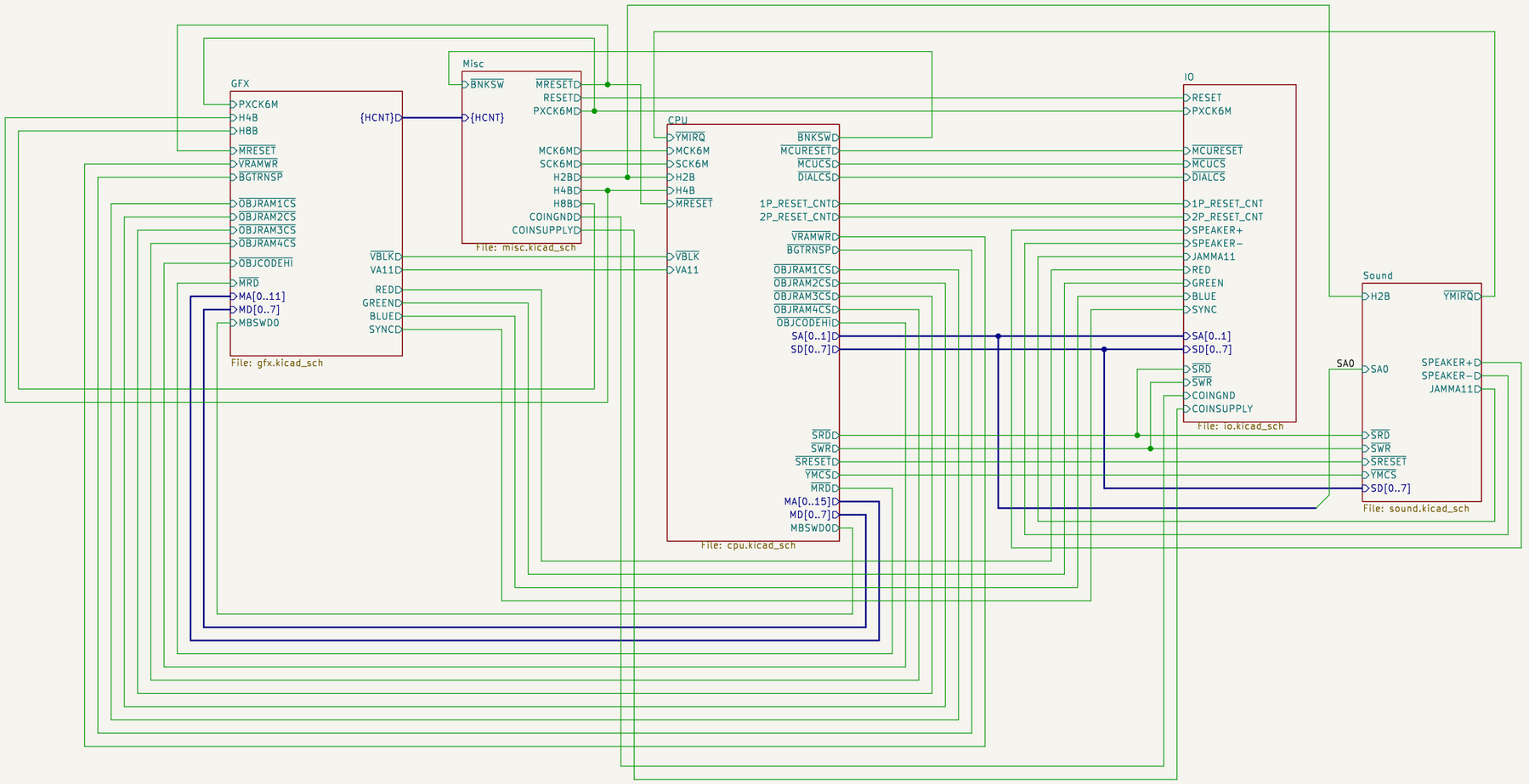


Seta number: P0-022-A
Ulf Skutnabba, twitter: @skutis77

Sheet: /
File: arknoid2.kicad_sch

Title: Arkanoid: Revenge of Doh

Size: A3	Date: 2023-02-22	**Rev:**
KiCad E.D.A. 8.0.5		Id: 1/6

E9:B06-10

-VRAMWR = MA15 & MA14 & MA13 & MA12 & ~MA11 & ~MA10 & ~MMREQ & MRFSH
-BGTRNSP = MA15 & MA14 & MA13 & MA12 & ~MA11 & MA10 & ~MA9 & ~MMREQ & MRFSH & ~MWR
-BNKSW = MA15 & MA14 & MA13 & MA12 & ~MA11 & MA10 & MA9 & ~MMREQ & MRFSH & ~MWR
-OBJCODELO = MA15 & MA14 & ~MA13 & ~MA12 & ~MMREQ & MRFSH
-OBJCODEHI = MA15 & MA14 & ~MA13 & MA12 & ~MMREQ & MRFSH
-MSHRAMRQ = MA15 & MA14 & MA13 & ~MA12 & ~MMREQ & MRFSH
-OBJORVRAMEN = MA15 & MA14 & MA13 & MA12 & ~MA11 & ~MMREQ & MRFSH & (~MA9 & ~MWR + ~MA10) + MA15 & MA14 & ~MA13 & ~MMREQ & MRFSH

D6:B06-11

-MROMCS = ~MMREQ & MRFSH & (~MA15 & ~MA14 & MR2A14 + ~MA14 & ~MR2A14 + ~MA15 & MA14)
-MROM2CS = MA15 & ~MA14 & ~MMREQ & MRFSH & MR2A14
MRA14 = MA15 & MBSWDO + ~MA15 & MA14
-OBJRAM4CS = VA11 & (OBJCODELO & H4B + ~OBJCODELO)
-OBJRAM3CS = ~VA11 & (OBJCODELO & H4B + & ~OBJCODELO)
-OBJRAM2CS = VA11 & (H4B + ~OBJCODEHI)
-OBJRAM1CS = ~VA11 & (H4B + ~OBJCODEHI)

C3:B06-12
i7 and i8 have been set to logical one in the equations.

SWAIT = YMCS2 & ~YMCS & (~SSHRAMEN + SSHRAMRQ)
MWAIT = SSHRAMEN & (OBJORVRAMEN + H2H4)
SSHRAMEN = SSHRAMRQ + ~MSHRAMRQ & SSHRAMEN
MSHRAMEN = MSHRAMRQ + ~SSHRAMRQ & MSHRAMEN
SHRAMCS = MSHRAMRQ & SSHRAMRQ
SHRAMWR = MWR & SSHRAMEN + MSHRAMRQ & SSHRAMRQ + SWR & MSHRAMEN
SHRAMENB = SSHRAMEN

C2:B06-13

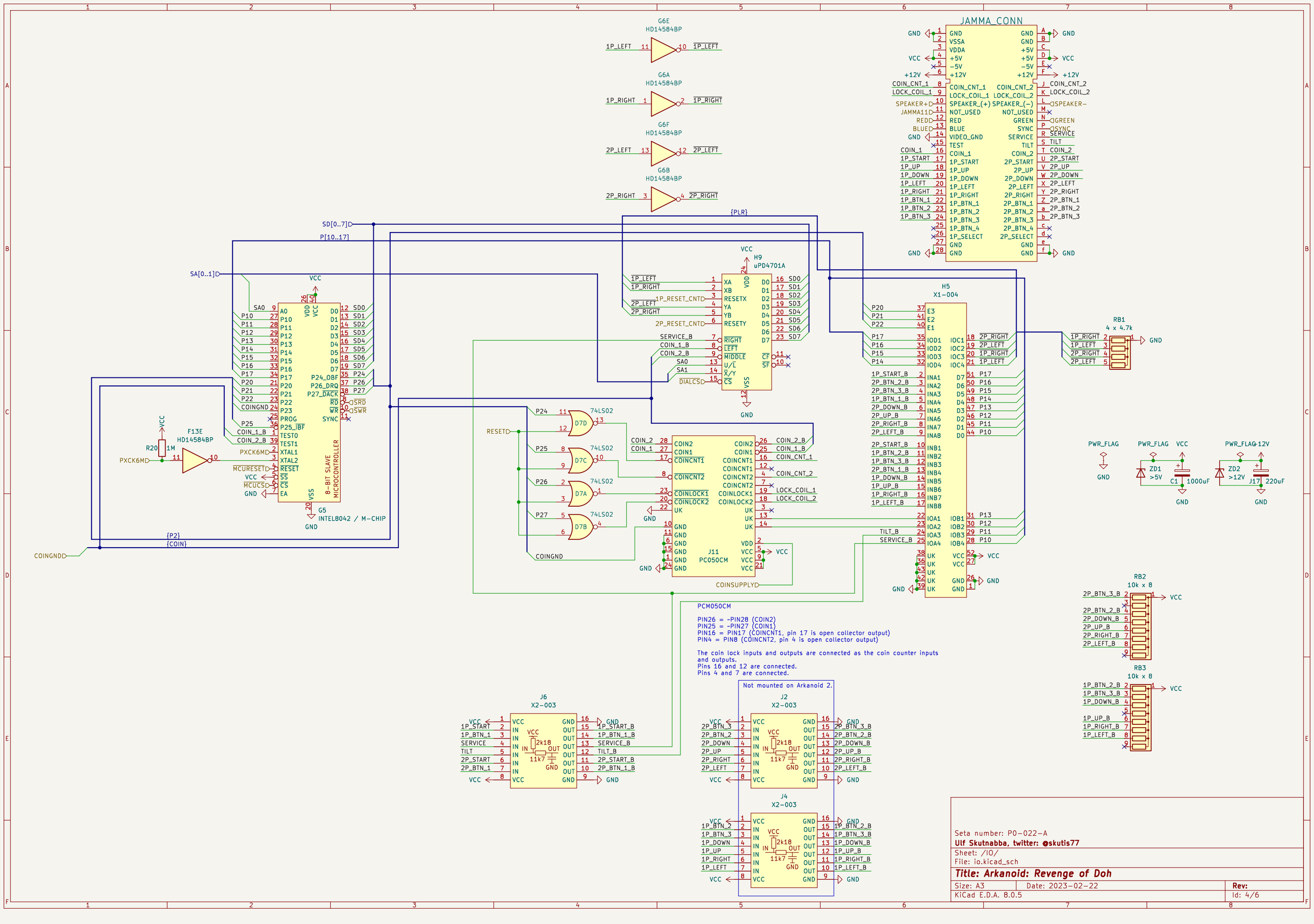
-SROMCS = ~SMREQ & SRFSH & (~SA14 & ~SA13 + ~SA15 & SA14 + ~SA15 & ~SA14)
-SSHRAMRQ = ~SMREQ & SRFSH & (SA15 & SA14 & SA13 & ~SA12 + SA15 & SA14 & ~SA13 & SA12)
-SBNKSW = SA15 & ~SA14 & SA13 & ~SA12 & ~SWR & ~SMREQ & SRFSH
-YMCS = SA15 & ~SA14 & SA13 & SA12 & ~SMREQ & SRFSH
-MCUCS = SA15 & SA14 & ~SA13 & ~SA12 & ~SMREQ & SRFSH
-DIALCS = SA15 & SA14 & SA13 & SA12 & ~SMREQ & SRFSH
SRA13 = SA15 & ~SBANK0 + ~SA15 & SA13
SRA14 = SA15 & ~SBANK1 + ~SA15 & SA14

BNKSW Also kicks watchdog
Ändra i insectx objorvramen till aktivt låg

Not mounted

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Sheet: /CPU/
File: cpu.kicad_sch
Title: Arkanoid: Revenge of Doh
Size: A3 | Date: 2023-02-22 | **Rev:**
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E10
74LS245
B6A
74LS08
CC6D
74F32
E11
74LS245
B6B
74LS08
CC6A
74F32
OSC1
OSC-12MHz
R4
330
R3
22
B17
YM3906
SETA_X1-001
B15
YM3909
SETA_X1-002
F17
MB7124
F15
MB7124
B08-07
B08-08
Color PROM
G17
8 x 470R
G16
8 x 470R
DAC
HH16
X1-003
RED
GREEN
BLUE
SYNC
H15
R0-012
H16
R0-012
H17
R0-012
SYNC and HBLK are probably
in a wired and configuration.
B13
74LS367
CC13
74LS367
3-StateHexBuffers
VCCD13
HM6116P-3
VCCD16
HM6116P-3
VCCD14
HM6116P-3
VCC D17
HM6116P-3
2kB RAM
B1D
74F04
CC6B
74F32
A15
74LS174
hex flip flop
74F04
A16
74LS273
flip flops
JP5
JP6
A13
27C512
A10
27C512
B08-01
A7
27C512
B08-02
A4
27C512
B08-03
A12
27C512
B08-04
A8
27C512
A5
27C512
A2
27C512
Not mounted
MD[0..7]
MA[0..11]
VD[0..15]
{VA}
{HCNT}
{CHA}
{GD}
{GE}
GA[5..15]
COL[0..8]
RGB[1..15]
GOE
GCE
GA15
PXCK6M
SYNC
VBLK
HBLK
VSYNC
BGTRNSP
OBJCODEHI
MRD
VRAMWR
MBSWD0
H4BD
H8BD
OBJRD
MRESET
OBJRAM1CS
OBJRAM2CS
OBJRAM3CS
OBJRAM4CS
HBLK:
f = 15.625kHz,
T = 64us,
Tactive = 53.33us,
Tblank = 10.67us
Total pixels / line = T * f_PXCLK = 64us x 6MHz = 384 pixels.
Active pixels / line = 53.33us x 6MHz = 320 pixels.
Blank pixels / line = 10.67us x 6Mhz = 64 pixels.
VSYNC:
f = 59.1856Hz
T = 16.896ms
Tactive = 16.384ms
Tblank = 0.512ms
Total lines = 264 lines
Active lines = 16.384ms x 15.625kHz = 256 lines
Blank lines = 8 lines.
5.31us = 32 pixels
2 pixels
2 pixels
SYNC is the wired and of HBLK and VSYNC.
SYNC is delayed by two pixels with respect to HBLK and VSYNC.
When VSYNC goes low, it is 5.31us after the rising edge of HBLK.
The delays are the same when VSYNC goes high, it is 5.31us after the
rising edge of HBLK.
Seta number: P0-022-A
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Sheet: /GFX/
File: gfx.kicad_sch
Title: Arkanoid: Revenge of Doh
Size: A3
Date: 2023-02-22
KiCad E.D.A. 8.0.5
Rev:
Id: 3/6



Seta number: PO-022-A
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Sheet: /IO/
File: io.kicad_sch

Title: Arkanoid: Revenge of Doh

Size: A3	Date: 2023-02-22	**Rev:**
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SCPU controls databus if it is floating or not.

SD[0..7]

SD0 SD1 SD2 SD3 SD4 SD5 SD6 SD7

SA0D

SRDD SWRD YMCSD SRESETD

H2BD

YMIRQ

VCC G9 YM2203 VSS

D0 D1 D2 D3 D4 D5 D6 D7 A0 RD WR CS IC PHM IRQ CHA CHB CHC PHS SH OPO IOA0 IOA1 IOA2 IOA3 IOA4 IOA5 IOA6 IOA7 IOB0 IOB1 IOB2 IOB3 IOB4 IOB5 IOB6 IOB7

SWA SW_DIP_x08

SWB SW_DIP_x08

GND

H11 YM3014

PH1 LOAD SD MP RB BUFF

H12B H12A RC4558

R7 1k

C9 10uF

C11 4.7uF

R8 6.2k

JAMMA11

C10 100nF

R9 8.2k

R21 1k

C12 10nF

VR1 890R

C13 1uF

C18 47uF

C17 47uF

C15 47uF

C16 47uF

C19 47uF

C14 47uF

R10 2.2

C20 100nF

R11 2.2

C21 100nF

SPEAKER+

SPEAKER−

J16 HA1388

18 W BTL Audio Power Amplifier

Amp1 Amp2

ASO_Protect Surge Protect TSD

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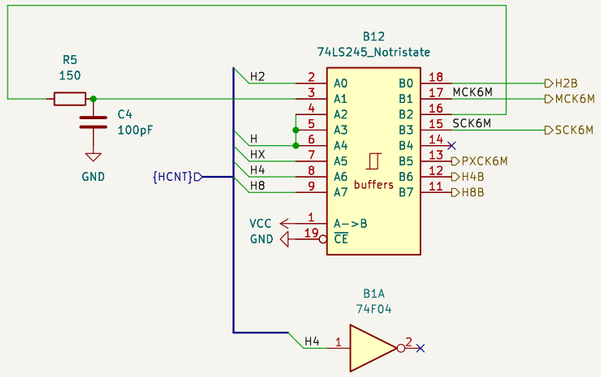
Sheet: /Sound/

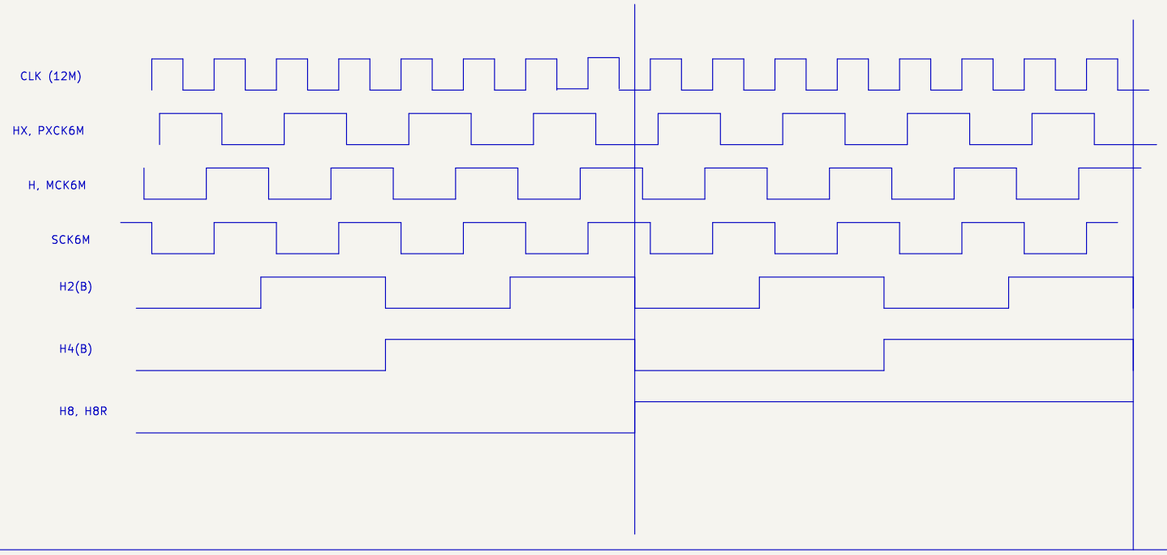
File: sound.kicad_sch

Title: Arkanoid: Revenge of Doh

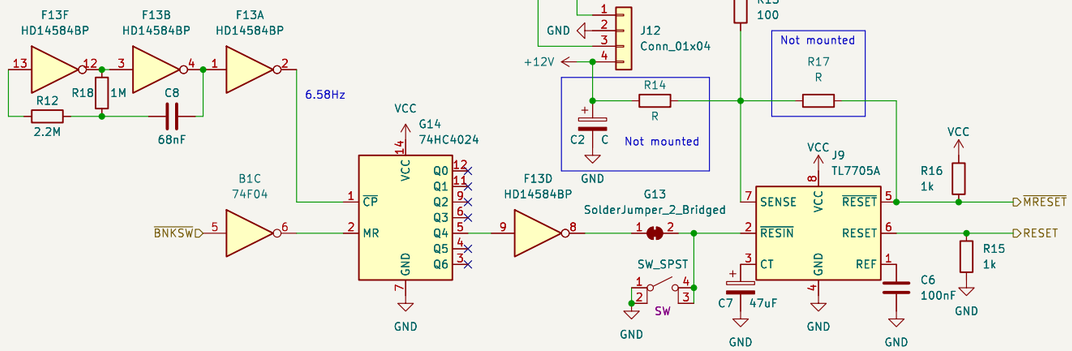
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Clocks





During normal operation pin 1 is shorted to pin 2 and pin 3 is shorted to pin 4.



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Sheet: /Misc/ File: misc.kicad_sch		
Title: Arkanoid: Revenge of Doh		
Size: A3	Date: 2023-02-22	**Rev:**
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